

PROJECT MSSA-80: EXECUTIVE BLUEPRINT

Subject: Conceptual Technical Feasibility and Economic Viability Analysis of a Next-Generation Mid-Size Supersonic Commercial Airliner Utilizing Advanced Composite Structural Architectures.

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1. Executive Summary

This report compiles the multi-disciplinary technical framework for a Mid-Size Supersonic Airliner (MSSA-80) configured for an 80-passenger payload capacity. Optimized for trans-oceanic North Atlantic corridors (Range: 4,200 nautical miles) operating at a sustained cruise velocity of Mach 1.7, the MSSA-80 breaks the economic and physical paradigms that hindered first-generation supersonic transport (SST). By systematically integrating modern Advanced Composite Materials (ACMs), non-afterburning variable-cycle propulsion, safety-enhanced anti-misting chemistry, and low-boom aerodynamics, the vessel achieves direct commercial parity with standard international Business Class yield profiles, projecting high financial sustainability at a 1.2× premium ticket pricing ceiling.

2. Economic Architecture & Yield Capture Model

The core business logic dictates the minimization of Cost per Available Seat Mile (**CASM**) via degressivity of scale, structural weight reduction, and asset utilization maximization. Traditional SST frameworks suffered from a polarization towards low capacities (20–30 seats) which drastically inflated fixed-cost distribution per passenger, or excessive volumes (120+ seats) which exponentially compounded wave drag scaling penalties.

2.1 Demand Capture Dynamics

The addressable market is calculated utilizing a specialized Yield Capture Equation focused on high-density premium corridors:

$$D_{ss} = M_{premium} \times \alpha \times [1 - (P_{ss} / P_{sub})^{-\epsilon}]$$

Where $M_{premium}$ represents the baseline daily subsonic premium traffic (modeled at 3,000 passengers/day for JFK→LHR), α is the time-saving capture coefficient (0.45 for a 3.5-hour flight duration), P_{ss} / P_{sub} is the target price ratio (1.2), and ϵ is the premium price elasticity parameter (−0.7). This model yields a stable capture rate of approximately 420–500 premium passengers per day, per direction, easily justifying multiple high-frequency fleet deployments.

2.2 Asset Utilization Metrics

Operating at Mach 1.7 reduces transatlantic flight time to 3.5 hours. Accounting for a strict 2-hour ground turnaround matrix for safety inspections, refueling, and catering, a single airframe executes up to 3 to 4 productive legs per day. This represents a 200% increase in capital asset utilization over a modern subsonic widebody airliner, effectively offsetting the reduced absolute seat capacity of the narrower airframe.

3. Advanced Composite Structural Mass Fraction

The physical viability of the aircraft relies entirely on replacing aluminum alloys with Advanced Composite Materials (ACMs). The legacy aluminum fuselage expanded significantly under the 127°C thermal skin load at Mach 2.04, requiring heavy mechanical expansion joints and thick structural reinforcement to handle high cabin-pressurization differentials at 60,000 feet.

3.1 Mass Breakdown and Fuel Fraction Allocation

Utilizing high-modulus Carbon Fiber Reinforced Polymers (CFRP) bound with high-temperature Bismaleimide (BMI) matrix resins reduces the Operating Empty Weight (W_{OE}) by 25–30% relative to aluminum equivalents. The structural weight savings trigger a cascading down-sizing effect across the engines, wing area, and fuel capacity requirements. The resulting vehicle mass budget is calculated via the structural mass-fraction equation:

$$W_{TO} = W_{OE} + W_{PL} + W_F$$

Mass Component Parameter	Legacy Aluminum Baseline	MSSA-80 Composite Metric	Operational Engineering Advantage
Payload Fraction (W_{PL})	6.0%	9.0%	50% increase in revenue-generating mass per unit of aircraft takeoff weight.
Operating Empty Fraction (W_{OE})	44.0%	36.0%	Anisotropic aeroelastic tailoring allows ultra-thin, low-drag, rigid wing profiles.
Fuel Mass Fraction (W_F)	50.0%	55.0%	Higher relative capacity allocated directly to mission range instead of lifting airframe deadweight.

4. Aerodynamics & Fuselage Cabin Geometry

4.1 Fuselage Shaping & Cabin Layout

To comply with the Sears-Haack body theory for wave drag minimization, the fuselage must preserve a highly slender cross-sectional area distribution. The internal space is optimized as a strict 1-1 single-aisle

configuration containing 40 rows. Deep-Cradle Ergonomic Recliners with a 42-inch seat pitch are used instead of heavy, space-consuming full lie-flat pods. This preserves passenger luxury for the shortened 3.5-hour duration while reducing individual seat weight by 60% and keeping the frontal cross-sectional area exceptionally narrow.

4.2 Wing Aerodynamics: Cranked-Arrow Delta Geometry

The aircraft utilizes a highly specialized Cranked-Arrow Delta wing to handle both supersonic cruise and subsonic low-speed airport maneuvers:

- **Inboard Sweep (72°):** Ensures that the leading edge of the wing root stays completely inside the conical shockwave (Mach Cone) generated by the nose at Mach 1.7, eliminating detached shockwave drag.
- **Outboard Sweep (53°):** Expands the outer span efficiency, ensuring stable aerodynamic control during low-speed subsonic airport approaches.
- **Vortex Lift Exploitation:** At high angles of attack during takeoff and landing, the aggressive inboard sweep generates strong, stable leading-edge vortices. This produces a powerful low-pressure field over the upper wing surface, delivering substantial non-linear lift and completely eliminating low-speed stalling.

5. Propulsion, Controls & Mishap-Resilient Fuel System

5.1 Variable-Cycle Engine (VCE) Architecture

The propulsion framework consists of twin non-afterburning Variable-Cycle Engines. By utilizing mechanical bypass valves, the engine dynamically adjusts its internal bypass ratio during flight:

Subsonic Airport Operations: Functions as a High-Bypass Turbofan. Diverts air around the core to reduce exhaust velocity, drastically decreasing decibel output to fully comply with strict international Chapter 14 airport noise limits.

Supersonic Cruise Mode (Mach 1.7): Transitions to a Low-Bypass Turbojet. Directs air entirely into the high-pressure core, generating a narrow, ultra-high-velocity exhaust stream capable of sustaining supersonic cruise speeds efficiently without fuel-heavy afterburners.

5.2 Safety-Enhanced Fuel Chemistry

The MSSA-80 integrates a specialized mishap-resilient, anti-misting fuel mixture. High-molecular-weight polymers are blended into a Sustainable Aviation Fuel (SAF) base, preventing the fuel from atomizing into a highly explosive mist during an uncontained structural rupture or impact. To allow combustion inside the VCE core, high-shear mechanical degraders are integrated into the engine feed lines. These devices use intense mechanical friction to break down the polymer chains milliseconds before injection, restoring volatile atomization only inside the isolated combustion chamber.

5.3 Navigation & Flight Control Solutions

- **Tailless Elevon System:** Pitch and roll control are managed by multi-segmented elevons on the wing trailing edges, actuated via a triple-redundant digital Fly-By-Wire system.
- **Center of Gravity (CoG) Fuel Management:** To counter the natural 6-foot backward shift of the Center of Pressure during supersonic transition, the flight computer automatically pumps the dense fuel mixture to an aft-fuselage trim tank, shifting the CoG without creating any trim drag.
- **Synthetic Vision Navigation:** To optimize structural integrity and aerodynamic profiling, the aircraft utilizes a rigid, non-drooping nose. External optical and radar sensors stream real-time, zero-delay data to a high-definition Synthetic Vision System (SVS) in the cockpit, providing pilots with exceptional visibility during landing.

6. Environmental & Regulatory Compliance

The aircraft is designed from the ground up to operate legally within modern environmental frameworks:

- **Overland Sonic Boom Mitigation:** Advanced slender fuselage shaping prevents shockwaves from coalescing into a standard double-boom. The shockwaves are diffused into an overland "sonic thump" measuring below 75 Perceived Loudness Decibels (PLdB), aligning perfectly with emerging noise-based performance standards.
- **Stratospheric Emissions Control:** Lean-Premixed Prevaporized (LPP) combustors maintain a uniform flame temperature inside the engine, reducing Nitrogen Oxide (NO_x) generation by up to 70% to protect the stratospheric ozone layer.
- **Net-Zero Compliance:** The composite fuel tanks are chemically lined with fluoropolymer coatings compatible with 100% Sustainable Aviation Fuel (SAF) bases, satisfying all CORSIA carbon emission standards.